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https://github.com/julianmak/OCES4303_ML_ocean

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 4303 :
an introduction to data-driven and ML methods in ocean sciences

Session 9: Convolutional Neural Networks

Outline

- ▶ basics of **PyTorch**
 - redoing last time's MLPs
- ▶ convolutions
 - **kernels** and **pooling**
 - resizing issues
 - examples
- ▶ **Convolutional Neural Networks (CNNs)**
 - TL;DR: kernel **weights** as part of control variables
 - classification and regression demonstration



Figure: Convolved ad-hoc TA.

Oceanic application

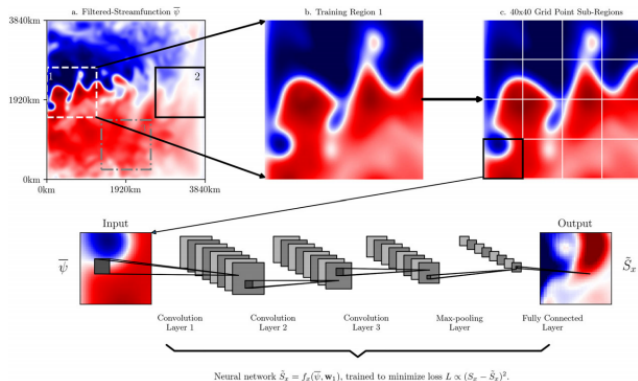


Figure: CNN applied to an eddy parameterisation problem. From Bolton & Zanna (2019).

► eddy parameterisation problem

→ regression problem: predict one image from another

Oceanic application

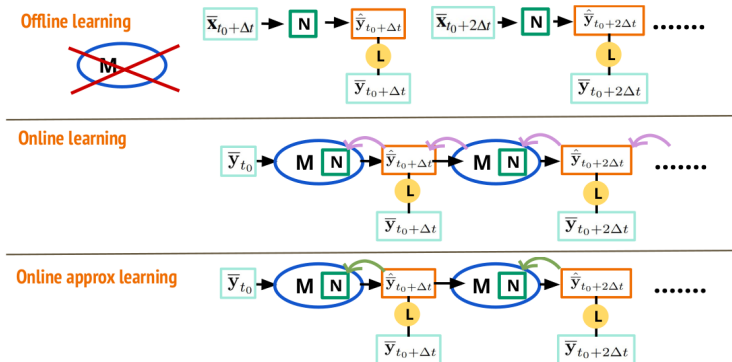


Figure: CNN training nested inside an eddy parameterisation problem. From Yan *et al.* (2025).

- as before, but nesting a CNN within a model (cf. RNN next time)
→ related to **variational data assimilation** approaches

PyTorch

- ▶ **PyTorch** and **TensorFlow** the two most popular libraries for neural networks
 - underlying them are procedure to manipulate **tensors** (multi-dim arrays)
 - links up to solvers and loss functions accordingly
 - have interfaces if need be (e.g. **Lightning** and **Keras**)
- ▶ **JAX** (e.g. w/ Keras) could be used in principle



PyTorch: building MLP as last time

- ▶ need to convert arrays into **tensor** objects
 - so PyTorch library knows what to do with them
 - extra useful functionalities (e.g. reshaping, resizing after manipulation)
- ▶ FloatTensor: **floats** (but `float32`)
- ▶ LongTensor: **signed integers**
- ▶ ByteTensor: **unsigned integers**
- ▶ in principle will do 'sane' things, but best be explicit
 - e.g. `torch.tensor([1., 1., 1.])` will default to FloatTensor
 - e.g. `torch.tensor([1, 1, 1])` will default to ByteTensor, will **fail** when passed to `conv1d`

PyTorch: building MLP as last time

- specify neural network structure

```
# define the model architecture

class nn_classify_pets(nn.Module):
    def __init__(self): # specify input dims below
        super(nn_classify_pets, self).__init__()

        # nest it in one go
        self.layers = nn.Sequential(
            nn.Linear(64*2, 100), # image is 64*2, 100 nodes (sklearn default)
            # nn.ReLU(),          # no activation function
            nn.Linear(100, 2)     # two possible outputs
        )

        # actual model structure: input -> hidden layer -> outputs, ReLU on input and hidden
    def forward(self, x):
        out = self.layers(x)
        return out
```

- define as a class, need `__init__`
- specify network structure
- need a `forward` to specify the feed-forward
→ this is for internal **taping/annotation** for doing back-propagation later

PyTorch: building MLP as last time

- ▶ define loss function
 - for binary classification we can choose `CrossEntropy`
 - others would work too
- ▶ define optimiser
 - use Adam again
 - need to initialise model (e.g. `model = MLP()`) and pass model parameters to the optimiser
 - telling the optimiser what are the control variables
- ▶ sequence of things to do:
 - do a feed-forward
 - evaluate loss function
 - back-propagate
 - update parameters, repeat

PyTorch: building MLP as last time

```
for epoch in range(num_epochs):  
  
    # iteration step  
  
    model.train() # put the model in training mode (taping is on)  
    optimizer.zero_grad() # clear gradients if it exists (from loss.backward())  
    Y_pred = model(X_train) # feed-forward  
    J_train = J(Y_pred, Y_train) # compute loss  
    J_train.backward() # back propagation  
    optimizer.step() # iterate  
    model.eval() # put the model in evaluation mode (taping is off for diags below)  
  
    # diagnostics: evaluation of metrics as we go along  
    with torch.no_grad(): # force no taping just in case  
        Y_pred = model(X_valid)  
        J_valid = J(Y_pred, Y_valid)  
        train_J[epoch] = J_train.item()  
        valid_J[epoch] = J_valid.item()
```

- ▶ above subroutine chains it all up, iterates over epochs
 - safety call to clear some calculations to avoid contamination
 - extra diagnostic calls

PyTorch: building MLP as last time

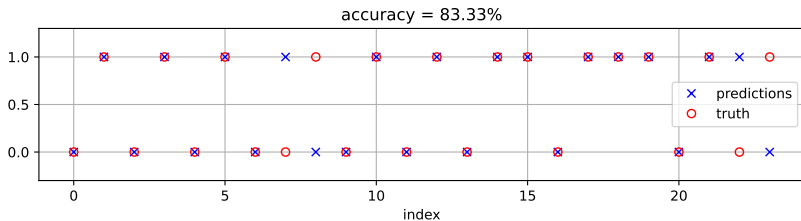
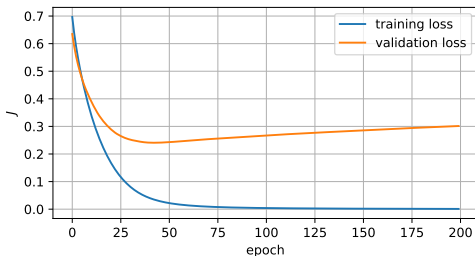


Figure: Perceptron with two hidden layers, no activation (cf. one of the cases considered last time but with `sklearn.MLPClassifier`).

CNNs

- ▶ **Convolutional Neural Networks (CNNs)**
 - has slightly better control on number of degrees of freedom compared to fully connected MLPs
 - naturally suited to images

CNNs

- **Convolutional Neural Networks (CNNs)**

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- naturally suited to images

- should be self-explanatory why after going through what **convolutions** are...

- recall from OCES 3301 where convolutions were used in relation to **filtering** in time series data:

$$(f * G)(x) = \int f(x')G(x, x') \, dx'$$

where instead of (x, x') we had (t, τ)

- easier in the discrete case with an example...

Convolutions

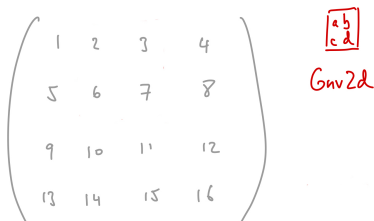


Figure: Sample array to be convolved with a kernel (the red stuff). Kernel here is of size (2,2).

Convolutions

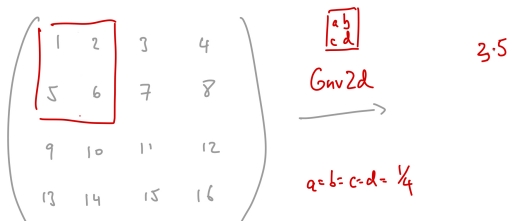


Figure: Choose a kernel, throw down kernel, element-wise multiplication, then sum.

Convolutions

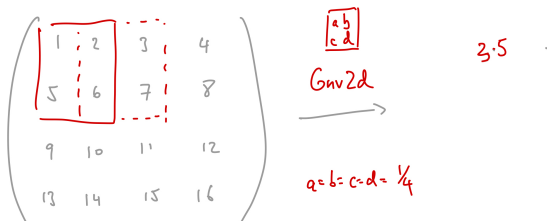


Figure: Move the kernel with some **stride** (1 here), then continue same process.

Convolutions

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix} \xrightarrow[\substack{\text{Conv2d} \\ a=b=c=d=\frac{1}{4}}]{\begin{matrix} a & b \\ c & d \end{matrix}} \begin{pmatrix} 3.5 & . & 5.5 \\ . & . & . \\ . & . & . \end{pmatrix}$$

Figure: Repeat until whole array is done. By default convolution leads to an array with **reduced** the size.

Convolutions

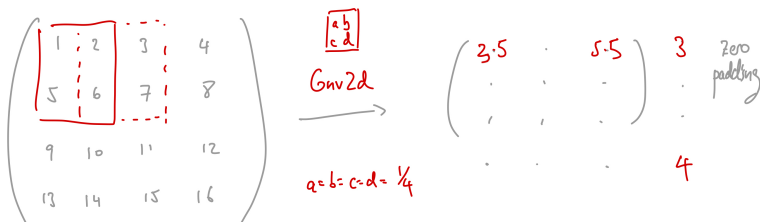


Figure: Can do **padding** for various reasons. Zero padding here bulks the **input** array size.

Pooling

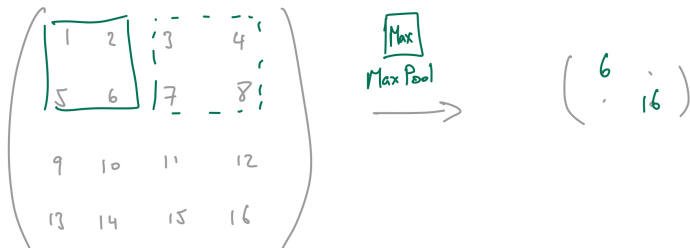


Figure: Pooling is similar, but default has stride equal to kernel size. This also reduces size of array.

- all the above can in principle be done for non-square kernels/poolings and/or unequal strides, paddings (and dilations; look that up if you want)

Convolution example

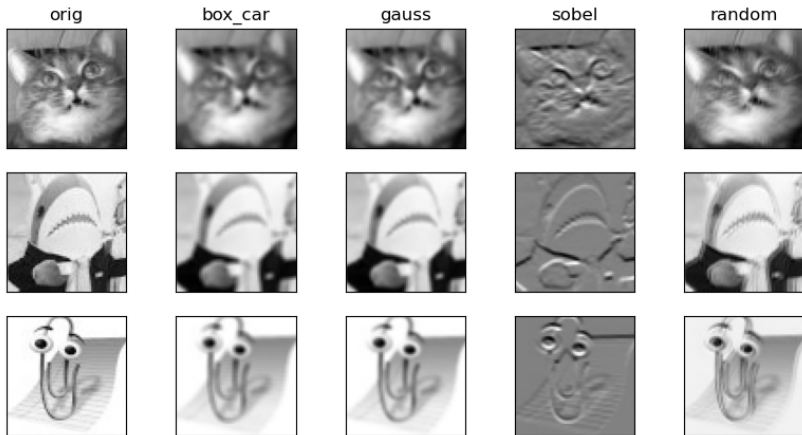


Figure: Convolving the ad-hoc TAs with a size 3 kernel with stride 1 with different choices of kernel weights. Input size is (64, 64) and output size in this case is (62, 62) (or slice 3-1 = 2 pixels off).

Pooling example

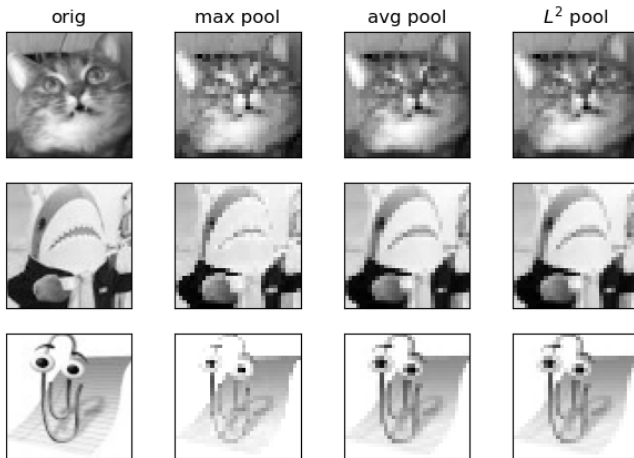


Figure: Pooling the ad-hoc TAs with a size 2 kernel with stride 2 with different operations. Input size is (64, 64) and output size in this case is (31, 31) (or divide by 2 in this case).

CNNs

- ▶ CNNs include **kernel weights** as control variables
 - otherwise proceed entirely as before
- ▶ controls degrees of freedom a bit better
 - number of kernel weights
 - reduction in array size
 - linear layer(s) as before

```
# convolutional layer 1 & max pool layer 1
self.layer1 = nn.Sequential(
    nn.Conv2d(1, 6, 5),      # now (6, 60, 60)
    nn.ReLU(),
    nn.MaxPool2d((2, 2)),    # now (6, 30, 30)
)

# convolutional layer 2 & max pool layer 2
self.layer2 = nn.Sequential(
    nn.Conv2d(6, 10, 5),     # now (10, 26, 26)
    nn.ReLU(),
    nn.MaxPool2d((2, 2)),    # now (10, 13, 13)
)

self.layer3 = nn.Sequential(
    nn.Linear(10 * 13 * 13, 60),
    nn.ReLU(),
    nn.Linear(60, 30),
    nn.ReLU(),
    nn.Linear(30, 2)         # because two possible
)

def forward(self, x):
    out = self.layer1(x)
    out = self.layer2(out)
    out = out.view(-1, 10 * 13 * 13)
    out = self.layer3(out)
    return out
```

Need some care in specifying sizes in the network structure!

CNNs: classification

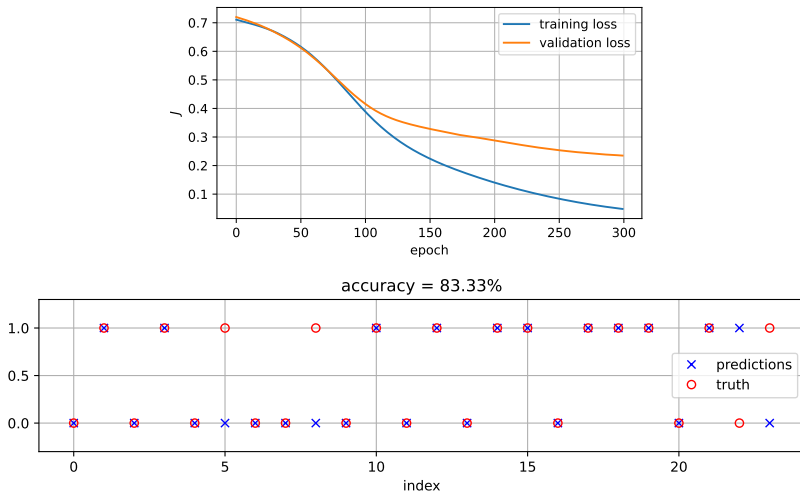


Figure: CNN for classification. Probably hasn't converged, but skill seems ok.

CNNs: regression

- ▶ need a change in the loss
→ use `MSELoss`
- ▶ need to change the CNN structure
→ linear layer to expand back to image array

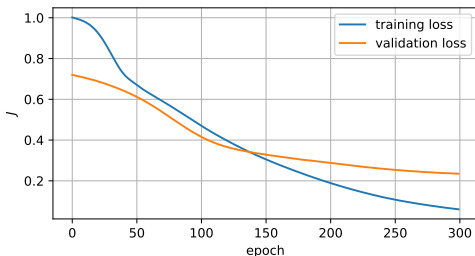


Figure: CNN for regression (predict bottom from top half).

CNNs: regression

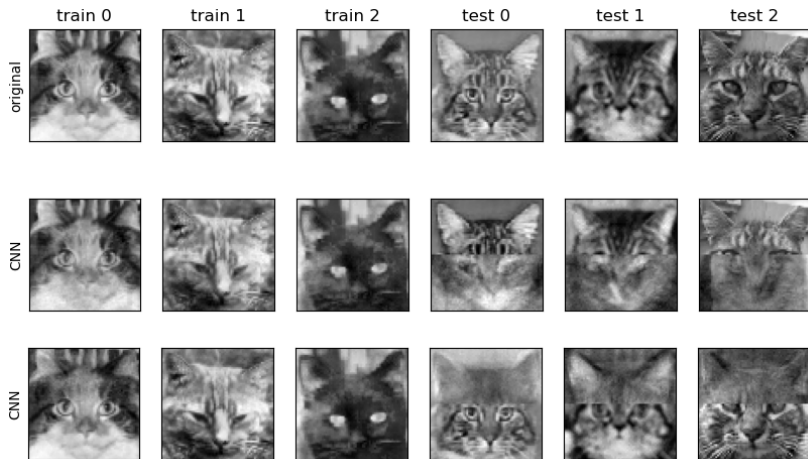


Figure: CNN for regression (predict bottom from top half and vice-versa).

Demonstration

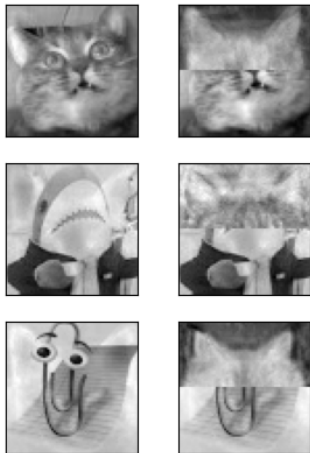


Figure: Grant us eyes (or not).

- ▶ introduced the use of PyTorch
→ did not use a interface (e.g. Keras), but you can try
- ▶ built a CNN
→ basics with layer structure etc.
→ need hyper-parameter tuning and cross-validation etc.

Moving to a Jupyter notebook →